

Weak lensing of the Lyman- α forest and its cross-correlation with CMB lensing: literature review and the sample-variance-limited estimator

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Abstract

These notes accompany an attempt to detect gravitational lensing of the DESI DR1 Lyman- α forest through its cross-correlation with CMB lensing maps from ACT and Planck. Section 1 reviews the (small) literature on lensing of the forest and the adjacent literature on forest–CMB-lensing correlations and on lensing reconstruction from three-dimensional source fields. Section 2 derives, in the flat-sky, single-source-plane, zero-noise (sample-variance) limit, the optimal quadratic estimator for the convergence from a set of forest pixels, the cubic estimator $W_{ijk}\delta_F^i\delta_F^j\delta_{F\kappa_{\text{CMB}}}^k$ for the cross-spectrum $C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}}$, its variance, the dense-sightline continuum limit with a closed-form reconstruction noise $N_\kappa(L)$, and a bias-hardened variant that projects out the response of the forest power spectrum to long-wavelength modes. Section 3 evaluates $N_\kappa(L)$ and the cross-correlation signal-to-noise for a fiducial forest power spectrum and DESI-like sightline densities. The central qualitative result is that for sparse sightlines most of the naive sample-variance information comes from an isotropic dilation of the transverse correlation function that is degenerate with a local modulation of the forest power spectrum amplitude, and that the robust (shear-like) information is set by the transverse rather than the line-of-sight resolution.

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1 Literature review

1.1 Lensing of the forest

Only one group has published on reconstructing the lensing potential from the forest itself. Croft et al. [1] proposed the idea: the three-dimensional forest is treated as a stack of source screens at $z \simeq 2\text{--}3$ and a configuration-space quadratic estimator, adapted from 21 cm lensing work, is applied to simulated spectra on a $15^\circ \times 15^\circ$ field with a sightline every 1.8 arcmin (which they call the extreme high end of achievable densities). They recover the degree-scale morphology of the potential and give no per-survey signal-to-noise. Metcalf et al. [2] supplied Gaussian noise forecasts. The noise per potential mode scales as $\sigma_\tau^2/[j_{\max}(l_{\max} - l_{\min})^2 l_{\min}^2]$, where σ_τ is the fractional per-pixel error and $j_{\max}$ the number of radial modes. Their cumulative signal-to-noise on the convergence power spectrum for DESI (770k spectra, 14 000 deg^2 , 8.1 arcmin mean separation, per-pixel S/N 3.3) is 7 for $\Delta z = 0.1$ source bins and 29 for $\Delta z = 0.5$ bins, corresponding to a 1.5 per cent measurement of the amplitude of mass fluctuations at $z \simeq 2.5$; eBOSS gives 4–16 and an MSE-like survey 50–160. Metcalf et al. [3] replaced the harmonic estimator by a real-space minimum-variance pixel-pair estimator built on the full pixel covariance, with the potential expanded in Legendre polynomials over a patch. This handles irregular sightline positions, holes and inhomogeneous noise. On Gaussian mocks they find that high-fidelity maps require $\gtrsim 0.5$ sources per arcmin^2 and signal-dominated pixels; a density of 200 deg^{-2} fails. Shaw et al. [4] applied that estimator to hydrodynamic simulations with ray-traced potentials: forest non-Gaussianity reduces the signal-to-noise by a factor of about 2.7 in the noise-free case and 1.5 for unit signal-to-noise spectra, and the recovered potential is biased by 5–25 per cent, which they correct by Gaussianizing the flux field. Shaw et al. [5] forecast DESI directly: 50 quasars per deg^2 , 700k spectra, pixel noise $\sigma_\delta = 0.3$ after rebinning to 2.8 Å and discarding the noisiest quarter, and detection defined as the correlation of the reconstructed potential with the foreground galaxy distribution. The full-survey significance is $S/N \simeq 3$ (two-pixel estimator), 4 (four-pixel) and 9 without pixel noise; 200–2000 deg^{-2} surveys on the same area reach 16–27. They state explicitly that they do not consider cross-correlation with CMB lensing.

No detection of forest lensing has been claimed, no analysis of eBOSS or DESI data exists, and no paper considers the cross-correlation of a forest-lensing reconstruction with CMB lensing. The last point is the gap this project targets: ACT DR6 [6, 7] and Planck PR4 [8] lensing maps cover the full DESI DR1 forest footprint of more than 9500 deg^2 with 428 403 valid Ly α -region forests [9], and a cross-correlation is immune to the additive Gaussian ($N^{(0)}$ -type) bias that dominates an auto-spectrum measurement.

1.2 Forest–CMB-lensing correlations that are not lensing of the forest

A separate line of work correlates the forest flux, or its local one-dimensional power, with κ_{CMB} . Vallinotto et al. [10, 11] forecast the two-point $\langle \delta_F \kappa_{\text{CMB}} \rangle$ and the $\langle \delta_F^2 \kappa_{\text{CMB}} \rangle$ correlations; Doux et al. [12] detected the latter at 5σ with BOSS DR12 and Planck 2015; Chiang and Slosar [13]

modelled it as the squeezed gravitational bispectrum plus a term from mean-flux misestimation on finite skewers; Karaçaylı et al. [14] detected it at 4.8σ with DESI Y1 forests and Planck PR3; La Posta and Schaan [15] forecast $S/N \simeq 10, 15$ and 20 for DESI with ACT, Simons Observatory and CMB-S4. Physically these measure the response of the small-scale forest power to long-wavelength modes, quantified with separate-universe simulations by Chiang et al. [16]. This is important here for a negative reason: as shown in Section 2.3, the same response mimics a lensing convergence in a quadratic estimator, so the detected $P_F \times \kappa_{\text{CMB}}$ bispectrum is the leading contaminant of the measurement we want to make. Loverde et al. [17] discuss the related but distinct effect of quasar magnification bias on forest statistics, at the 0.1–1 per cent level.

1.3 Lensing reconstruction from three-dimensional fields

The formalism below follows the CMB quadratic estimator of Hu [18], Hu and Okamoto [19] generalized to a three-dimensional source by Zahn and Zaldarriaga [20], who showed that each line-of-sight Fourier mode acts as an independent screen and that the screens combine with inverse-variance weights. Lu and Pen [21] pointed out that source non-Gaussianity sets a noise floor even without instrumental noise. Pourtsidou and Metcalf [22, 23] developed the 21 cm intensity-mapping version at $z \simeq 2\text{--}3$ including discreteness noise. Foreman et al. [24] computed the bias to the three-dimensional estimator from the matter bispectrum, found it to cost a factor of a few in signal-to-noise for 21 cm surveys, and introduced a bias-hardened estimator; Schaan et al. [25], Schaan and Ferraro [26] generalized the estimator to weakly non-Gaussian sources and introduced the shear-only estimator that is immune to isotropic modulations of the source power. Bias hardening in the CMB context is due to Namikawa et al. [27]; see also Sailer et al. [28], Maniyar et al. [29]. Buncher et al. [30] recently applied the standard and shear-only estimators to projected galaxy clustering as the source, the closest analogue of a sparse-tracer reconstruction. A fuller annotated list is kept in `report/literature_notes.md`.

2 Formalism in the sample-variance limit

2.1 Set-up and conventions

We work in the flat-sky, distant-observer approximation. A point in the forest volume is labelled by an angular position $\boldsymbol{\theta}$ (radians) and a comoving line-of-sight distance χ ; the effective comoving distance to the forest is χ_* ($\chi_* \simeq 3956 h^{-1}\text{Mpc}$ at $z = 2.4$ for a Planck-2018 cosmology). The flux contrast $\delta_F(\boldsymbol{\theta}, \chi)$ is treated as a statistically homogeneous Gaussian random field with the anisotropic (redshift-space) power spectrum $P_F(k_{\parallel}, k_{\perp})$. Its Fourier transform in the mixed angular/radial variables is

$$\delta(\boldsymbol{l}, k_{\parallel}) = \int d^2\theta d\chi \delta_F(\boldsymbol{\theta}, \chi) e^{-i(\boldsymbol{l}\cdot\boldsymbol{\theta} + k_{\parallel}\chi)}, \quad \langle \delta(\boldsymbol{l}, k_{\parallel}) \delta(\boldsymbol{l}', k'_{\parallel}) \rangle = (2\pi)^3 \delta_{\text{D}}^2(\boldsymbol{l} + \boldsymbol{l}') \delta_{\text{D}}(k_{\parallel} + k'_{\parallel}) \mathcal{P}(\boldsymbol{l}, k_{\parallel}), \quad (1)$$

with $\mathcal{P}(\boldsymbol{l}, k_{\parallel}) = P_F(k_{\parallel}, k_{\perp} = l/\chi_*)/\chi_*^2$. Reality of δ_F implies $\delta(-\boldsymbol{l}, -k_{\parallel}) = \delta^*(\boldsymbol{l}, k_{\parallel})$. For a survey of solid angle Ω and radial depth D the coincident delta functions are $(2\pi)^2 \delta_{\text{D}}^2(0) = \Omega$ and $2\pi \delta_{\text{D}}(0) = D$.

Lensing by a single effective source plane at χ_s is described by a projected potential $\phi(\boldsymbol{\theta})$ with deflection $\boldsymbol{\alpha} = \nabla\phi$, convergence $\kappa = -\nabla^2\phi/2$ (so $\kappa(\boldsymbol{L}) = L^2\phi(\boldsymbol{L})/2$) and shear $\gamma_1 = -(\partial_1^2 - \partial_2^2)\phi/2$, $\gamma_2 = -\partial_1\partial_2\phi$, exactly the convention of Hu and Okamoto [19]. The observed forest field is

$$\tilde{\delta}(\boldsymbol{\theta}, \chi) = \delta_F(\boldsymbol{\theta} + \nabla\phi(\boldsymbol{\theta}), \chi). \quad (2)$$

Because the source plane is common to all χ , the displacement is independent of the radial coordinate; this is the single approximation that makes the problem separable in $k_{\parallel}$. In reality the lensing kernel changes by a few per cent across a forest, which we defer to later work.

2.2 Mode coupling induced by lensing

Expanding (2) to first order, $\tilde{\delta} = \delta_F + \nabla\phi \cdot \nabla_{\theta}\delta_F$, and Fourier transforming,

$$\tilde{\delta}(\mathbf{l}, k_{\parallel}) = \delta(\mathbf{l}, k_{\parallel}) - \int \frac{d^2 l_1}{(2\pi)^2} [\mathbf{l}_1 \cdot (\mathbf{l} - \mathbf{l}_1)] \phi(\mathbf{l} - \mathbf{l}_1) \delta(\mathbf{l}_1, k_{\parallel}). \quad (3)$$

For a fixed realization of ϕ the two-point function of the observed field acquires an off-diagonal part. With $\mathbf{L} = \mathbf{l} + \mathbf{l}'$ and $\mathbf{L} \neq 0$,

$$\langle \tilde{\delta}(\mathbf{l}, k_{\parallel}) \tilde{\delta}(\mathbf{l}', k'_{\parallel}) \rangle_{\phi} = 2\pi \delta_D(k_{\parallel} + k'_{\parallel}) \left[(2\pi)^2 \delta_D^2(\mathbf{L}) \mathcal{P}(l, k_{\parallel}) + f(\mathbf{l}, \mathbf{l}'; k_{\parallel}) \phi(\mathbf{L}) \right], \quad f(\mathbf{l}, \mathbf{l}'; k_{\parallel}) = \mathbf{L} \cdot \mathbf{l} \mathcal{P}(l, k_{\parallel}) + \mathbf{L} \cdot \mathbf{l}' \mathcal{P}(l', k'_{\parallel}) \quad (4)$$

It is convenient to express the response in terms of the convergence, $f_{\kappa} \equiv 2f/L^2$, so that the off-diagonal term reads $f_{\kappa} \kappa(\mathbf{L})$. Equation (4) is diagonal in $k_{\parallel}$: every radial Fourier mode is an independently lensed two-dimensional screen with power spectrum $\mathcal{P}(l, k_{\parallel})$, all sharing the same ϕ [20]. Pairs with $k'_{\parallel} \neq -k_{\parallel}$ carry no lensing information and, for a Gaussian field, no covariance either, so an optimal estimator never uses them.

The pair-displacement picture. The same result follows from the real-space statement in the task description: a pair of pixels $i = (a, p)$ and $j = (b, q)$ on sightlines a, b at radial positions χ_p, χ_q has covariance

$$C_{ij}(\phi) = \xi(\boldsymbol{\theta}_{ab} + \boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b, \chi_p - \chi_q) \simeq \xi(\boldsymbol{\theta}_{ab}, \Delta\chi_{pq}) + \nabla_{\theta} \xi(\boldsymbol{\theta}_{ab}, \Delta\chi_{pq}) \cdot [\nabla\phi(\boldsymbol{\theta}_a) - \nabla\phi(\boldsymbol{\theta}_b)], \quad (5)$$

where ξ is the correlation function of δ_F in angular-radial coordinates and $\boldsymbol{\theta}_{ab} = \boldsymbol{\theta}_a - \boldsymbol{\theta}_b$. Only the relative deflection of the two members of a pair matters, so a pair measures the local tidal field of ϕ rather than the deflection itself. Fourier transforming (5) over both positions reproduces (4); this was verified symbolically for a periodic one-dimensional toy model in `mathematica/pair_displacement_check.wls`.

2.3 Squeezed limit: dilation, shear and the amplitude degeneracy

For lens modes much larger than the forest modes, $L \ll l$, expanding (4) to second order in L gives (verified in `mathematica/lensing_response_checks.wls`, check A)

$$f_{\kappa}(\mathbf{l}, \mathbf{L} - \mathbf{l}) \simeq 2\mathcal{P}(l) + 2(\hat{\mathbf{L}} \cdot \hat{\mathbf{l}})^2 l \mathcal{P}'(l) = \underbrace{[2\mathcal{P} + l\mathcal{P}']}_{\text{dilation}} + \underbrace{l\mathcal{P}' \cos 2\varphi_{lL}}_{\text{shear}}, \quad (6)$$

with $\mathcal{P}' = \partial\mathcal{P}/\partial l$ at fixed $k_{\parallel}$. The two pieces have a clean geometric meaning. For a pair separation θ_{ab} small compared with the lens wavelength, $\boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b \simeq (\boldsymbol{\theta}_{ab} \cdot \nabla) \nabla\phi = -(\kappa \mathbb{K} + \boldsymbol{\gamma}) \boldsymbol{\theta}_{ab}$, so the pair correlation is evaluated at the true separation $(1 - \kappa)\boldsymbol{\theta}_{ab} - \boldsymbol{\gamma} \boldsymbol{\theta}_{ab}$: the convergence isotropically dilates the transverse correlation function while the shear makes it anisotropic. In Fourier space the exact power spectrum of a field dilated and sheared by constant $\kappa, \gamma_1, \gamma_2$ is $\mathcal{P}(|M^{-T}\mathbf{l}|)/\det M$ with $M = \mathbb{K} - \kappa\mathbb{K} - \boldsymbol{\gamma}$, whose first-order expansion equals (6) (check B in the same script).

The amplitude degeneracy is now apparent. A slowly varying multiplicative modulation of the field, $\tilde{\delta} = (1 + a)\delta_F$, produces the coupling

$$\langle \tilde{\delta}(\mathbf{l}, k_{\parallel}) \tilde{\delta}(\mathbf{l}', k'_{\parallel}) \rangle \supset 2\pi \delta_D(k_{\parallel} + k'_{\parallel}) f_a(\mathbf{l}, \mathbf{l}'; k_{\parallel}) a(\mathbf{L}), \quad f_a = \mathcal{P}(l, k_{\parallel}) + \mathcal{P}(l', k_{\parallel}) \xrightarrow{L \ll l} 2\mathcal{P}, \quad (7)$$

so the dilation term $2\mathcal{P} + l\mathcal{P}' = (2 + n)\mathcal{P}$, with $n = d \ln \mathcal{P} / d \ln l$, is proportional to the amplitude response. For a transverse spectrum that is locally a power law the convergence is therefore *indistinguishable* from a local change of the forest power spectrum amplitude by $\delta \ln P_F = (2 + n)\kappa$; only the shear term $l\mathcal{P}' \cos 2\varphi$ and departures from a power law break the degeneracy. For $n = -2$

the dilation response vanishes identically, and for a spectrum that is flat in l ($n = 0$, which is what the forest looks like transversely whenever $l/\chi_* \ll k_\parallel$) there is no shear response at all and the entire lensing signal is an amplitude change. This matters because the forest power spectrum really is modulated by long-wavelength density and tidal modes, with responses measured by Chiang et al. [16], and those long modes are correlated with κ_{CMB} . That correlation is exactly the $P_F \times \kappa_{\text{CMB}}$ bispectrum detected by Doux et al. [12], Karaçaylı et al. [14], and it will enter our cubic estimator as an additive bias unless projected out (Section 2.7).

2.4 The optimal quadratic estimator for discrete pixels

Let $d_i = \tilde{\delta}(\theta_{a(i)}, \chi_{p(i)})$, $i = 1 \dots N_{\text{pix}}$, be the vector of all forest pixels on all sightlines, in the zero-noise limit. Expand the potential in a set of real basis functions, $\phi(\theta) = \sum_\mu p_\mu \psi_\mu(\theta)$; in the continuum limit below the ψ_μ are Fourier modes, while on a finite irregular footprint one may use Legendre or Zernike polynomials as in Metcalf et al. [3]. To first order in p the pixel covariance is, from (5),

$$C_{ij}(p) = C_{ij}^0 + \sum_\mu p_\mu C_{ij}^{\mu}, \quad C_{ij}^{\mu} = \nabla_\theta \xi(\theta_{ab}, \Delta\chi_{pq}) \cdot [\nabla \psi_\mu(\theta_a) - \nabla \psi_\mu(\theta_b)]. \quad (8)$$

For a Gaussian likelihood $-2 \ln \mathcal{L} = d^T C^{-1} d + \ln \det C$ the Newton–Raphson step from $p = 0$ is the optimal (minimum-variance, unbiased to first order) quadratic estimator,

$$\hat{p}_\mu = \frac{1}{2} \sum_\nu F_{\mu\nu}^{-1} \left[d^T C_0^{-1} C^{\nu} C_0^{-1} d - \text{Tr}(C_0^{-1} C^{\nu}) \right], \quad F_{\mu\nu} = \frac{1}{2} \text{Tr}(C_0^{-1} C^{\mu} C_0^{-1} C^{\nu}). \quad (9)$$

Writing $\hat{p}_\mu = \sum_{ij} E_{ij}^{\mu} d_i d_j - b_\mu$ identifies the quadratic weights and the mean field,

$$E^{\mu} = \frac{1}{2} \sum_\nu F_{\mu\nu}^{-1} C_0^{-1} C^{\nu} C_0^{-1}, \quad b_\mu = \sum_{ij} E_{ij}^{\mu} C_{ij}^0. \quad (10)$$

Using $\langle d_i d_j \rangle = C_{ij}(p)$ one checks $\langle \hat{p}_\mu \rangle = p_\mu + O(p^2)$, and for Gaussian d the covariance at $p = 0$ is $\text{Cov}(\hat{p}_\mu, \hat{p}_\nu) = F_{\mu\nu}^{-1}$, which saturates the Cramér–Rao bound. The convergence coefficients follow by applying $-\nabla^2/2$ to the basis; for Fourier modes $\hat{\kappa}_L = (L^2/2)\hat{\phi}_L$. Two remarks. First, the same formulae hold with pixel noise by replacing $C^0 \rightarrow C^0 + N$, so nothing in this section is specific to the sample-variance limit except the numerical value of F . Second, because E^μ is built from ratios of C^0 and its derivative, an overall rescaling of the forest power spectrum leaves the estimator and its variance unchanged: in the zero-noise limit the reconstruction noise depends only on the *shape* of P_F and on the geometry of the pixel set.

2.5 The cubic cross-correlation estimator

Let m_k be the pixels of a CMB convergence map and $\kappa_L^{\text{CMB}} = \sum_k Y_{Lk} m_k$ its (masked, apodized) flat-sky harmonic transform, normalized so that $\langle \kappa_L \kappa_{L'}^* \rangle = \Omega \delta_{LL'} C_L$ for a full-sky-normalized spectrum. Choosing Fourier modes as the basis in (9) and converting the potential weights to convergence weights, $E^{\kappa, L} = (L^2/2) E^{\phi, L}$ and $b_L^\kappa = (L^2/2) b_L^\phi$, the natural bandpower estimator for the cross-spectrum in a bin b containing N_b modes on a periodic patch is

$$\hat{C}_b = \frac{1}{\Omega N_b} \sum_{L \in b} \text{Re}[\hat{\kappa}_L \kappa_L^{\text{CMB}*}] = \sum_{ijk} W_{ijk}^b d_i d_j m_k - \frac{1}{\Omega N_b} \sum_{L \in b} \text{Re}[b_L \kappa_L^{\text{CMB}*}], \quad W_{ijk}^b = \frac{1}{\Omega N_b} \sum_{L \in b} \text{Re}[E_{ij}^{\kappa, L} Y_{Lk}^*]. \quad (11)$$

This is the cubic form $W_{ijk} \delta_F^i \delta_F^j \kappa_{\text{CMB}}^k$ anticipated in the task description, with the weights fixed by (10). On a masked sky the harmonic coefficients are not diagonal and the bandpower expectation must be corrected with the mask mixing (MASTER) matrix; (11) is the periodic-patch form. Its properties:

- **Expectation.** To first order in ϕ , $\langle \hat{C}_b \rangle = \bar{C}_b^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}}$, the bin average of the true cross-spectrum. The mean-field term is a fixed function of the pixel geometry and is uncorrelated with κ_{CMB} , so it averages to zero but should be subtracted to avoid excess variance.
- **No Gaussian ($N^{(0)}$) bias.** The disconnected part of $\langle d_i d_j m_k \rangle$ vanishes because $\langle m_k \rangle = 0$ and the Gaussian part of $d_i d_j$ is independent of m_k . This is the main advantage of the cross-correlation over the auto-spectrum of $\hat{\kappa}$, whose $N^{(0)}$ bias equals the full reconstruction noise.
- **Higher-order lensing corrections.** For Gaussian ϕ and a forest independent of ϕ , the $O(\phi^2)$ term of $\langle \tilde{\delta} \tilde{\delta} \kappa_{\text{CMB}} \rangle$ contains $\langle \phi \phi \kappa_{\text{CMB}} \rangle$, which vanishes, so there is no $N^{(1)}$ -type bias at second order in the cross-correlation. The leading Gaussian corrections to the cross-response appear at third order in the remapping expansion, schematically $C^{\phi\phi} C^{\phi\kappa_{\text{CMB}}}$, and a second-order term arises only from the lensing bispectrum or from non-Gaussian correlations between the forest and the lenses. These have not been computed here.
- **Intrinsic clustering bias.** δ_F is itself a biased tracer of the matter at $z \simeq 2-3$, which lies inside the CMB lensing kernel. The connected bispectrum $\langle \delta_F \delta_F \kappa_{\text{CMB}} \rangle_c$ therefore contributes even in the absence of lensing of the forest. Its squeezed part is the response of P_F to long modes, $\delta \ln P_F = R_\delta \delta_L + R_K K_L$ (density and tidal responses), correlated with κ_{CMB} ; in the language of Section 2.3 it enters as $\langle \hat{a}_L \kappa_L^{\text{CMB}*} \rangle \neq 0$ projected onto $\hat{\kappa}$ through the response ratio $F_{\kappa a}/F_{\kappa \kappa}$. This is the forest analogue of the gravitational-nonlinearity bias of Foreman et al. [24] and it is not small: it is the signal that Karaçaylı et al. [14] detected at 4.8σ in DESI Y1.
- **Variance.** For Gaussian fields and optimal weights the variance in the continuum limit is

$$\text{Var}(\hat{C}_b) = \frac{(C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} + N_\kappa(L))(C_L^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}} + N_L^{\text{CMB}}) + (C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}})^2}{f_{\text{sky}}(2L+1)\Delta L}, \quad (12)$$

with $N_\kappa(L)$ the reconstruction noise derived next. For a general pixel set the numerator's first factor is replaced by the appropriate combination of F^{-1} , i.e. by $\text{Cov}(\hat{\kappa}_L, \hat{\kappa}_{L'})$ from (9).

2.6 Continuum limit: Hu–Okamoto form and reconstruction noise

For infinitely dense sightlines and complete radial coverage over a depth D , the field is measured on a regular three-dimensional grid and the Fisher matrix of Fourier modes becomes diagonal. It is then simpler to derive the estimator directly. Take

$$\hat{\kappa}(\mathbf{L}) = A(L) \int \frac{dk_\parallel}{2\pi} \int \frac{d^2 l}{(2\pi)^2} g(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_\parallel) \tilde{\delta}(\mathbf{l}, k_\parallel) \tilde{\delta}(\mathbf{L} - \mathbf{l}, -k_\parallel). \quad (13)$$

Inserting (4), the coincident radial delta function gives $2\pi\delta_D(0) = D$, so $\langle \hat{\kappa}(\mathbf{L}) \rangle = \kappa(\mathbf{L})$ requires

$$A(L)^{-1} = D \int \frac{dk_\parallel}{2\pi} \int \frac{d^2 l}{(2\pi)^2} g f_\kappa. \quad (14)$$

The Gaussian variance of (13) at $\phi = 0$ has two Wick pairings, each producing another factor D and the power spectra of the two legs; minimizing it subject to the normalization gives the standard weight $g = f_\kappa/[2\mathcal{P}(l, k_\parallel)\mathcal{P}(|\mathbf{L} - \mathbf{l}|, k_\parallel)]$ and

$$\boxed{\frac{1}{N_\kappa(L)} = D \int_{-\infty}^{\infty} \frac{dk_\parallel}{2\pi} I_{\kappa\kappa}(L, k_\parallel), \quad I_{XY}(L, k_\parallel) = \int \frac{d^2 l}{(2\pi)^2} \frac{f_X(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_\parallel) f_Y(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_\parallel)}{2\mathcal{P}(l, k_\parallel)\mathcal{P}(|\mathbf{L} - \mathbf{l}|, k_\parallel)}} \quad (15)$$

with $\langle \hat{\kappa}(\mathbf{L}) \hat{\kappa}^*(\mathbf{L}') \rangle = (2\pi)^2 \delta_D^2(\mathbf{L} - \mathbf{L}') [C_L^{\kappa\kappa} + N_\kappa(L)]$. Equation (15) is the inverse-variance sum over independent $k_\parallel$ screens of the Hu–Okamoto noise of each screen, $N_{k_\parallel}^{-1} = I_{\kappa\kappa}$, with the mode density $D dk_\parallel / 2\pi$ counting the screens [20]. It is the same object as F^{-1} of (9) per unit solid angle. Two consistency checks: (i) since $f_\kappa \propto \mathcal{P}$, the integrand is invariant under $\mathcal{P} \rightarrow c\mathcal{P}$, as it must be in the sample-variance limit; (ii) for a white spectrum, $\mathcal{P} = \text{const}$, $f_\kappa = 2\mathcal{P}$ exactly, so $I_{\kappa\kappa} = l_{\text{max}}^2 / 2\pi$ and $N_\kappa^{-1} = D k_{\text{max}} l_{\text{max}}^2 / 2\pi^2$; this equals twice the number of real degrees of freedom per steradian of the field, which is what one obtains directly by noting that a dilation by κ changes the variance of every independent white-noise pixel by a factor $(1+2\kappa)$ and each pixel then carries Fisher information $\frac{1}{2}(2)^2 = 2$. The numerical implementation (`code/recon_noise.py`) reproduces this count to the accuracy of the $|\mathbf{L} - \mathbf{l}| \leq l_{\text{max}}$ edge exclusion.

The cutoffs l_{max} and $k_{\parallel, \text{max}}$ are where the real survey enters even in this idealization: transverse modes are only measured up to the sightline Nyquist frequency, $l_{\text{max}} \simeq \pi / \theta_{\text{sep}} = \pi \sqrt{n_q}$ for n_q sightlines per steradian, radial modes up to a fraction of the pixel Nyquist frequency, and radial modes below $k_{\parallel, \text{min}} \sim 0.03 h \text{ Mpc}^{-1}$ are removed by continuum fitting. With sightlines of mean forest length D_f covering only part of the total depth, the product $D l_{\text{max}}^2$ that controls (15) is fixed by the total sightline length per unit area, so one may use $D = D_f$ together with the full n_q . Sparse sampling also aliases small-scale transverse power into the measured band; that effect is beyond the continuum treatment and is one reason the numbers below are optimistic.

2.7 Hardening against amplitude modulation

An estimator $\hat{a}(\mathbf{L})$ for the amplitude modulation is built from (13) with f_a in place of f_κ , and the two estimators respond to each other's field with response functions $R_{\kappa a}(L) = F_{\kappa a} / F_{\kappa\kappa}$ and $R_{a\kappa} = F_{\kappa a} / F_{aa}$, where $F_{XY} = D \int dk_\parallel / 2\pi I_{XY}$. The bias-hardened estimator [27]

$$\hat{\kappa}^{\text{BH}} = \frac{\hat{\kappa} - R_{\kappa a} \hat{a}}{1 - R_{\kappa a} R_{a\kappa}}, \quad N_\kappa^{\text{BH}}(L) = \frac{N_\kappa(L)}{1 - \rho_{\kappa a}^2(L)}, \quad \rho_{\kappa a}^2 = \frac{F_{\kappa a}^2}{F_{\kappa\kappa} F_{aa}}, \quad (16)$$

has zero response to any $a(\boldsymbol{\theta})$ that multiplies all $k_\parallel$ modes equally. Since the true responses $R_\delta(k_\parallel, k_\perp)$ of the forest are scale dependent, a more conservative choice hardens each $k_\parallel$ screen separately, $1/N_\kappa^{\text{BH, slice}} = D \int dk_\parallel / 2\pi I_{\kappa\kappa} (1 - \rho_{\kappa a}^2(L, k_\parallel))$; this removes any modulation of the form $a(\boldsymbol{\theta}, k_\parallel)$ and keeps only the information that is genuinely anisotropic or shape-changing within a screen. Both variants are evaluated below; the truth for the real forest lies in between and also requires hardening against the tidal response, which mimics shear rather than convergence and would need the $k_\perp$ -dependence of R_K to be separated from $l\mathcal{P}'$.

2.8 The signal and its detectability

In the Limber approximation, with a single forest source plane at χ_* and the CMB at χ_{CMB} ,

$$C_L^{XY} = \int_0^{\chi_{\text{min}}} d\chi \frac{W_X(\chi) W_Y(\chi)}{\chi^2} P_m\left(k = \frac{L + 1/2}{\chi}, z(\chi)\right), \quad W_s(\chi) = \frac{3}{2} \Omega_m \left(\frac{H_0}{c}\right)^2 (1+z) \chi \frac{\chi_s - \chi}{\chi_s}, \quad (17)$$

and the total detection significance of the cross-spectrum from (12) is

$$\left(\frac{S}{N}\right)^2 = \sum_L f_{\text{sky}} (2L + 1) \frac{(C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}})^2}{(C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} + N_\kappa(L)) (C_L^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}} + N_L^{\text{CMB}}) + (C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}})^2}. \quad (18)$$

Because the forest source plane sits well inside the CMB lensing kernel the two convergences are strongly correlated, with $r_L = C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}} / (C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} C_L^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}})^{1/2} \simeq 0.9$ at $L = 10$ falling to 0.67 at $L \gtrsim 300$.

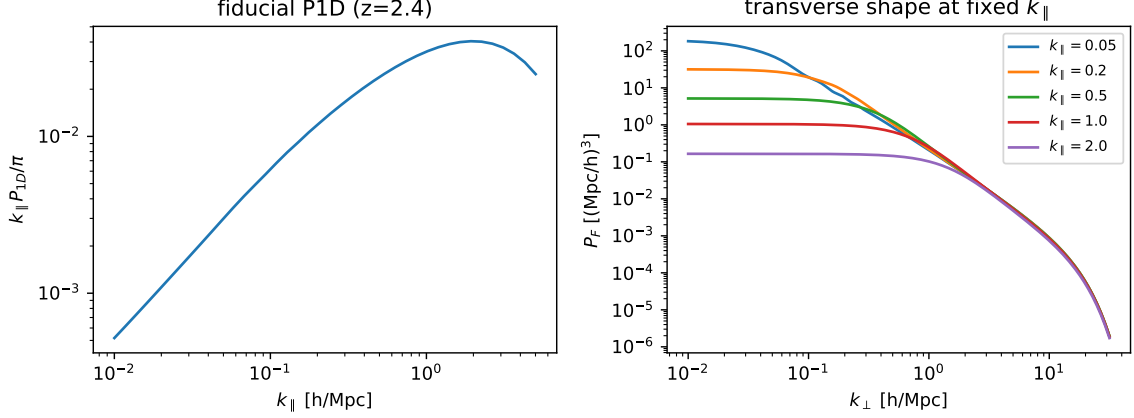


Figure 1: Fiducial forest power spectrum at $z = 2.4$. Left: the implied one-dimensional power. Right: the transverse shape at fixed $k_{\parallel}$; for $k_{\perp} \ll k_{\parallel}$ the spectrum is flat in l , which is the regime in which lensing is degenerate with an amplitude change (Section 2.3).

3 Toy numbers

Fiducial model. The forest power spectrum is the Kaiser form with the non-linear correction of Arinyo-i-Prats et al. [31], $P_F = b_F^2(1 + \beta_F \mu^2)^2 P_{\text{lin}}(k) D_{\text{NL}}(k, \mu)$, with approximate $z = 2.4$ parameters $b_F = -0.13$, $\beta_F = 1.6$, $q_1 = 0.6$, $q_2 = 0$, $k_v = 1 \text{ h Mpc}^{-1}$, $a_v = 0.55$, $b_v = 1.6$, $k_p = 16 \text{ h Mpc}^{-1}$ (code/forest_power.py; only the shape matters here). The linear and non-linear matter power spectra come from CAMB for a Planck-2018 cosmology. The radial depth is $D = 350 \text{ h}^{-1} \text{ Mpc}$ (one forest), the radial band is $0.03 \leq k_{\parallel} \leq k_{\parallel, \text{max}}$ with $k_{\parallel, \text{max}} \in \{1, 2, 5\} \text{ h Mpc}^{-1}$, and the transverse cutoff is the sightline Nyquist multipole for $n_q \in \{20, 50, 100, 400\} \text{ deg}^{-2}$ ($l_{\text{max}} = 1273$ for the DESI-DR1-like value $n_q = 50$; DR1 has about 45 forests per deg^2 over 9500 deg^2). Figure 1 shows the fiducial spectrum.

CMB lensing noise. Rather than fabricate noise curves, we use white noise levels calibrated to reproduce the published detection significances of the auto-spectra: $N_L^{\text{CMB}} = 1.68 \times 10^{-7}$ reproduces 43σ over $40 \leq L \leq 763$ on $f_{\text{sky}} = 0.23$ for ACT DR6 [6], and $N_L^{\text{CMB}} = 3.03 \times 10^{-7}$ reproduces the $\simeq 42\sigma$ -equivalent amplitude error over $8 \leq L \leq 400$ on $f_{\text{sky}} = 0.67$ for Planck PR4 [8]. Real noise curves rise steeply at high L and should replace these before any forecast is quoted externally. Overlap fractions of $f_{\text{sky}} = 0.15$ (ACT) and 0.22 (Planck, the full DESI DR1 forest area) are assumed.

Results. Figure 2 compares $N_{\kappa}(L)$ with the signal spectra and Table 1 lists values at $L = 100$, where $C_{100}^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} = 2.78 \times 10^{-8}$, $C_{100}^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}} = 4.57 \times 10^{-8}$ and $C_{100}^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}} = 1.38 \times 10^{-7}$.

Three features of these numbers are robust even though their absolute values are not:

1. For a DESI-like density the naive sample-variance noise at $L \sim 100$, $N_{\kappa} = 2.45 \times 10^{-8}$, is comparable to the signal $C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}}$, and it falls steeply with $k_{\parallel, \text{max}}$ (as $1/k_{\parallel, \text{max}}$ when the transverse spectrum is flat) and with n_q (as $1/l_{\text{max}}^2 \propto 1/n_q$).
2. The amplitude-hardened noise is ~ 8 times larger for $n_q = 50$ ($N_{\kappa}^{\text{BH}} = 1.98 \times 10^{-7}$), ~ 17 times larger for $n_q = 20$, and it is almost independent of $k_{\parallel, \text{max}}$. This is the quantitative form of Section 2.3: radial modes with $k_{\parallel} \gg l_{\text{max}}/\chi_*$ see a transversely flat spectrum, and for them lensing is purely an amplitude change. The information that survives hardening comes from the few screens with $k_{\parallel} \lesssim l_{\text{max}}/\chi_* \simeq 0.3 \text{ h Mpc}^{-1}$, so it is set by the sightline density alone. Increasing the density from 50 to 400 deg^{-2} reduces N_{κ}^{BH} by a factor ~ 30 ,

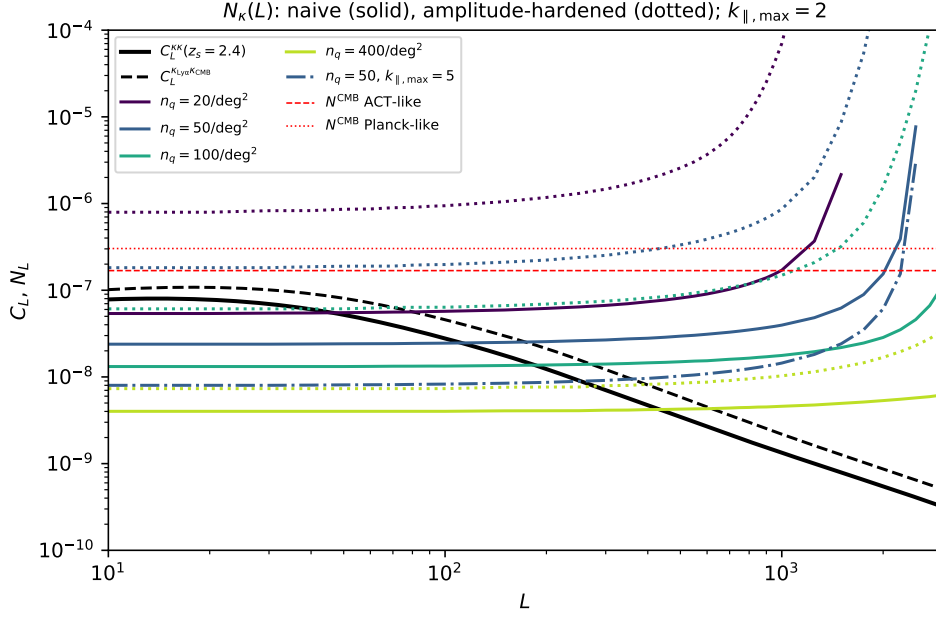


Figure 2: Sample-variance-limited reconstruction noise $N_\kappa(L)$ (solid, Eq. 15) for sightline densities n_q and $k_{\parallel,\max} = 2 h \text{ Mpc}^{-1}$, its per-screen amplitude-hardened version (dotted, Eq. 16), and the case $k_{\parallel,\max} = 5$ for $n_q = 50$ (dash-dotted). Black curves are the forest convergence auto-spectrum and its cross-spectrum with κ_{CMB} ; red lines are the white-noise equivalents of the ACT DR6 and Planck PR4 reconstruction noise.

Table 1: Reconstruction noise at $L = 100$ and cumulative cross-correlation signal-to-noise (to $L = 3000$) in the sample-variance limit. N^{BH} is the per-screen amplitude-hardened noise. S/N columns are for $k_{\parallel,\max} = 2$; the last column uses N^{BH} with ACT-like CMB noise.

$n_q [\text{deg}^{-2}]$	$l_{\max}$	$k_{\parallel,\max}$	$N_\kappa(100)$	$N_\kappa^{\text{BH}}(100)$	S/N ACT	S/N Planck	S/N ACT, BH
20	805	1.0	1.36×10^{-7}	9.49×10^{-7}			
		2.0	5.71×10^{-8}	9.44×10^{-7}	25.7	25.4	7.1
		5.0	2.04×10^{-8}	9.43×10^{-7}			
50	1273	1.0	6.21×10^{-8}	2.01×10^{-7}			
		2.0	2.45×10^{-8}	1.98×10^{-7}	36.5	35.7	15.2
		5.0	8.27×10^{-9}	1.97×10^{-7}			
100	1800	1.0	3.40×10^{-8}	6.63×10^{-8}			
		2.0	1.33×10^{-8}	6.37×10^{-8}	46.1	44.6	25.0
		5.0	4.26×10^{-9}	6.32×10^{-8}			
400	3600	1.0	8.15×10^{-9}	8.84×10^{-9}			
		2.0	4.01×10^{-9}	7.33×10^{-9}	67.4	64.2	54.4
		5.0	1.21×10^{-9}	7.05×10^{-9}			

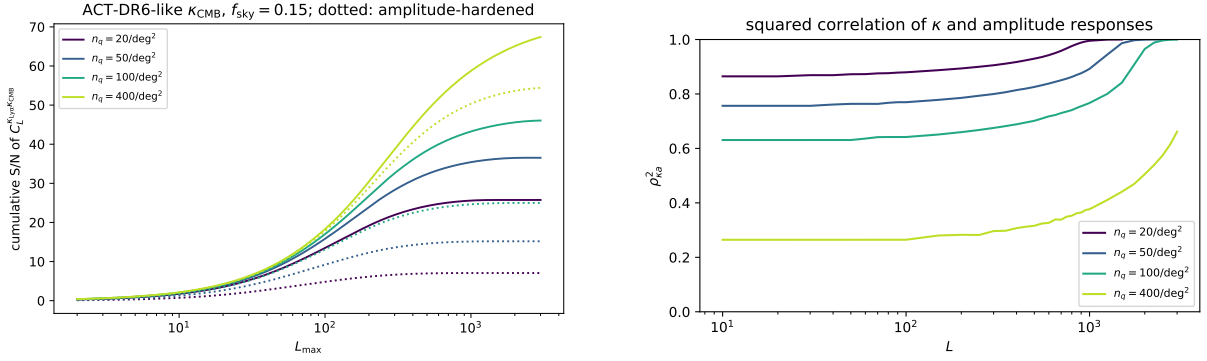


Figure 3: Left: cumulative signal-to-noise of $C_L^{K_{Ly\alpha} K_{CMB}}$ for ACT-like CMB noise and $f_{sky} = 0.15$, with (dotted) and without (solid) amplitude hardening. Right: squared correlation coefficient between the convergence and amplitude responses, ρ_{ka}^2 of Eq. (16), after summing over $k_{||}$ screens.

faster than the n_q^{-1} scaling of the naive noise, because it also converts amplitude-degenerate modes into shear-sensitive ones.

3. With ACT-like CMB noise the sample-variance-limited cross-correlation would be detected at $\simeq 37\sigma$ for $n_q = 50$ (Planck: 36σ), dominated by $L \lesssim 500$. These numbers are upper limits of no practical relevance by themselves: DESI pixel noise ($\sigma_\delta \simeq 0.8$ per $0.8 \text{ \AA}$ pixel against a forest rms of 0.34) removes most radial modes above $k_{||} \sim 0.5 h \text{ Mpc}^{-1}$, sparse sampling aliases transverse power, and the estimator's own $N^{(1)}$ and clustering biases are not included. Shaw et al. [5] find $S/N \simeq 3\text{--}4$ for the reconstructed potential against foreground galaxies with realistic noise. The route from the numbers here to a realistic forecast is to add the pixel noise to $\mathcal{P}$ in (15), treat the sightline sampling explicitly through (9), and compute the bias terms.

4 Three tracers: separating lensing from the forest response

Section 3 showed that hardening the estimator against an amplitude modulation costs most of the sample-variance information. This section asks a different question: with three fields on the same footprint, the quasars q at the forest redshift, the forest δ_F and κ_{CMB} , which power spectra and bispectra separate the lensing of the forest from its response to long-wavelength density and tidal modes without assuming that response is known? The answer is formulated in the squeezed limit and at the level of spectra only.

4.1 Squeezed-limit model

Let $\hat{P}_F(\mathbf{k}|\boldsymbol{\theta})$ be the forest power spectrum measured locally in a cell around $\boldsymbol{\theta}$ over the redshift slab of interest (in practice the pixel-pair estimator of Section 2.4 restricted to a cell, or the position-dependent power spectrum). Its fractional fluctuation, $p(\mathbf{k};\boldsymbol{\theta}) = \delta \ln \hat{P}_F(\mathbf{k}|\boldsymbol{\theta})$, is a two-dimensional field for every short-mode bin $\mathbf{k} = (k_{||}, k_{\perp}, \varphi)$, with φ the angle of $\mathbf{k}_{\perp}$ to the long-mode direction $\hat{\mathbf{L}}$. To first order in the long modes,

$$p(\mathbf{k}; \mathbf{L}) = T(\mathbf{k}) \kappa_{Ly\alpha}(\mathbf{L}) + \mathcal{R}(\mathbf{k}) \delta_s(\mathbf{L}) + \dots, \quad (19)$$

where δ_s is the matter density projected over the slab and:

- $T(\mathbf{k}) = (2 + n_{\perp}) + n_{\perp} \cos 2\varphi$, with $n_{\perp} = \partial \ln P_F / \partial \ln k_{\perp}$ at fixed $k_{||}$, is the lensing template of Eq. (6). It is *known*, because it is built from the measured P_F itself.

- $\mathcal{R}(\mathbf{k}) = R_\delta(k_\parallel, k_\perp) + R_K(k_\parallel, k_\perp) \cos 2\varphi$ is the response of the forest to the long mode. For a long mode with $k_\parallel \simeq 0$, which is the only kind that projects onto a two-dimensional field such as κ_{CMB} , the tidal tensor is fixed by the density, $K_{ij} = (\hat{L}_i \hat{L}_j - \delta_{ij}/3) \delta_s$, so the tidal response is not an independent field: it is a $\cos 2\varphi$ harmonic multiplying the same $\delta_s(\mathbf{L})$ (`mathematica/three_tracer_checks.wls`, check 5). $\mathcal{R}$ is treated as *unknown*: one number per short-mode bin and harmonic.

Both terms have the same angular structure (an $m = 0$ and an $m = 2$ harmonic), so they cannot be separated by the angular dependence alone. They can be separated because they multiply different long-mode fields: $\kappa_{\text{Ly}\alpha}$ is sourced by matter at $z < z_s$, δ_s by matter at $z \simeq z_s$, and the quasars trace the latter and not the former.

The quasar overdensity projected over the same slab is

$$q(\mathbf{L}) = b_q \delta_s(\mathbf{L}) + m \kappa_q(\mathbf{L}) + \epsilon(\mathbf{L}), \quad m \equiv 5s - 2, \quad (20)$$

with $b_q \simeq 3.5$ the quasar bias, $m \kappa_q$ the magnification-bias term ($\kappa_q \simeq \kappa_{\text{Ly}\alpha}$ since the quasars sit at the forest redshift) and ϵ shot noise plus any stochasticity.

4.2 The measurements and the deprojection identity

Cross-correlating p with the two long-mode fields gives two families of bispectra (they are $\langle \delta_F \delta_F X \rangle$ in squeezed configurations, with the short modes binned in $\mathbf{k}$):

$$B_{FF\kappa}(\mathbf{k}; L) \equiv \langle p(\mathbf{k}) \kappa_{\text{CMB}} \rangle_L = \mathcal{R}(\mathbf{k}) C_L^{\delta_s \kappa_{\text{CMB}}} + T(\mathbf{k}) C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}}, \quad (21)$$

$$B_{FFq}(\mathbf{k}; L) \equiv \langle p(\mathbf{k}) q \rangle_L = \mathcal{R}(\mathbf{k}) C_L^{\delta_s q} + T(\mathbf{k}) C_L^{\kappa_{\text{Ly}\alpha} q}. \quad (22)$$

B_{FFq} is the quasar–forest-power cross-correlation proposed by Chiang et al. [16]; $B_{FF\kappa}$ integrated over $\mathbf{k}_\perp$ is the P_{1D} – κ_{CMB} bispectrum of Doux et al. [12], Karaçaylı et al. [14], for which T vanishes identically because a single skewer is invariant under a transverse remapping (the dilation response integrates to a boundary term, $\int_0^{l_{\text{max}}} (2\mathcal{P} + l\mathcal{P}') l dl = l_{\text{max}}^2 \mathcal{P}(l_{\text{max}})$, check 4). Form the linear combination

$$\mathcal{D}(\mathbf{k}; L) \equiv B_{FF\kappa}(\mathbf{k}; L) - \beta_L B_{FFq}(\mathbf{k}; L) = T(\mathbf{k}) \left[C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}} - \beta_L C_L^{\kappa_{\text{Ly}\alpha} q} \right], \quad \beta_L \equiv \frac{C_L^{\delta_s \kappa_{\text{CMB}}}}{C_L^{\delta_s q}} \quad (23)$$

in which the unknown response cancels identically for every $\mathbf{k}$ (check 1). Equivalently, one cross-correlates the full quadratic estimator $\hat{\kappa}$ of Section 2.6 with the deprojected CMB map $\kappa_\perp \equiv \kappa_{\text{CMB}} - \beta q$.

Kernel matching. Equation (23) holds under the model (19), in which the response multiplies a single projected density. In the measurement the response of a short-mode bin is a radial integral $p_\alpha^{\text{int}} = \int d\chi A_\alpha(\chi) \delta(\chi)$ with bin-dependent weights A_α set by forest evolution, coverage, noise and continuum fitting. The residual after deprojection is then

$$\int d\chi A_\alpha(\chi) \left[W_{\text{CMB}}(\chi) - \beta_L b_q(\chi) W_q(\chi) \right] \frac{P((L + \frac{1}{2})/\chi, \chi)}{\chi^2}, \quad (24)$$

which vanishes for arbitrary A_α only if the bracket vanishes pointwise in χ . A single slab-averaged β with a top-hat quasar kernel does not achieve this: W_{CMB} rises from 3.61×10^{-4} to $4.86 \times 10^{-4} h \text{ Mpc}^{-1}$ across the $2.1 < z < 3.0$ slab (`mathematica/pair_template_checks.wls`, check 8). The fix is to build the quasar template with a radial weight that matches the CMB kernel,

$$\hat{\kappa}_s(\boldsymbol{\theta}) = \int_{\text{slab}} d\chi W_{\text{CMB}}(\chi) \frac{\delta_q(\boldsymbol{\theta}, \chi)}{b_q(\chi)} = \sum_{i \in \boldsymbol{\theta}} \frac{W_{\text{CMB}}(\chi_i)}{b_q(\chi_i) \bar{n}_{3D}(\chi_i) \chi_i^2} - \text{mean}, \quad (25)$$

an estimate of the contribution of the slab itself to κ_{CMB} . Then $\beta \equiv 1$, the bracket in (24) vanishes identically for any radial response weight, and the only inputs are $W_{\text{CMB}}(\chi)$ (geometry) and $b_q(z)$, which DESI measures to a few per cent from the quasar auto-correlation. A fractional error ϵ in b_q leaves a fraction ϵ of the contamination: the residual is first order in the coefficient error, not second order. The shot noise of the weighted template is $D^2 \langle W_{\text{CMB}}^2 \rangle / (b_q^2 \bar{n}_q) = 8.99 \times 10^{-8}$, marginally above the $\beta^2 / \bar{n}_q = 8.58 \times 10^{-8}$ of the single-coefficient version.

What still has to hold.

1. **Locality in χ .** The forest response must be to the density at the same χ ; a long mode with $k_{\parallel} \neq 0$ within the slab window ($k_{\parallel} \lesssim 1/D$) has a tidal tensor that differs from the transverse one at $O(k_{\parallel}^2/k^2)$ and a quasar redshift-space distortion at $O(\mu^2)$; both are suppressed for the $k_{\parallel} \lesssim 0.01 h \text{ Mpc}^{-1}$ modes that project onto a two-dimensional field, but the residual radial transfer mismatch between density, tides and quasar RSD must be tested on mocks rather than assumed.
2. **Deterministic quasar bias.** Stochasticity beyond Poisson noise leaves $(1 - r_{q\delta}^2)$ of the contamination (check 2); Poisson noise adds variance, not bias.
3. **Magnification.** With $q = b_q \delta + m \kappa_q$, the template (25) contains $(m/b_q) \int W_{\text{CMB}} \kappa_q$, a lensing term. In the single-coefficient version the exact cancellation coefficient is $\beta = C^{\delta_s \kappa_{\text{CMB}}} / (b_q C^{\delta_s \delta_s} + m C^{\delta_s \kappa_q})$ and ignoring m altogether biases β by $\Delta\beta/\beta = 9$ per cent at $L = 100$ for $m = 0.5$ (check 3b), which leaks first order into the residual. $m = 5s - 2$ must therefore be fitted or measured, most cleanly from C_L^{qq} with a low-redshift galaxy sample, which has no intrinsic overlap. Matter inside the slab also lenses the more distant forest pixels, so the template subtracts real signal: the total subtraction $C^{\kappa_{\text{Ly}\alpha} \kappa_s} / C^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}}$ is 1.1 per cent at $L = 100$ in the fiducial model, and it is kept in the signal template rather than neglected.
4. **Shared quasars.** The template quasars in a slab also provide forest pixels in the lower part of the same slab, so the local sightline density is correlated with q . Taking pixels only from quasars behind the whole slab would remove standard-region coverage below $z \simeq 2.4$ and is not an option. The measurement should instead use narrow tomographic sub-slabs with disjoint template and sightline selections, monitor the sightline-density $\times \kappa_{\text{CMB}}$ cross-spectrum, and model the shared sampling in mocks.

The complete set of measurements is therefore: (i) $B_{FF\kappa}(\mathbf{k}; L)$, (ii) $B_{FF\hat{\kappa}_s}(\mathbf{k}; L)$ with the matched template, (iii) C_L^{qq} in redshift shells for $b_q(z)$, (iv) C_L^{qq} with a low- z galaxy sample for magnification, and (v) the forest correlation function that defines the lensing template. The lensing detection is the amplitude of $\mathcal{D}$ against the template; the consistency test is that $\mathcal{D}/T$ is independent of $\mathbf{k}$; and $B_{FF\hat{\kappa}_s}$ measures the response $\mathcal{R}$ itself, up to the small magnification-lensing leak. Section 5 implements the same logic without cells or harmonics.

4.3 Realistic noise

The sparse, noisy sampling of the forest enters (15) by replacing $\mathcal{P} \rightarrow \mathcal{P} + \mathcal{N}$ in the denominators, with the response f_{κ} unchanged (self-pairs of a sightline carry no lensing response, Eq. 8 with $a = b$, and only add variance). For Poisson-distributed sightlines with effective density $\bar{n}_{\text{eff}}$ per steradian at the redshift in question, the noise is white in l and given by the one-dimensional power of a skewer,

$$\mathcal{N}(k_{\parallel}) = \frac{P_N + P_{1D}(k_{\parallel})}{\bar{n}_{\text{eff}}}, \quad P_N = \sigma_{\delta}^2 \Delta\chi_{\text{pix}}, \quad (26)$$

the form used for three-dimensional forest power spectra by McQuinn & White. The P_{1D} term is the aliasing of unresolved transverse power; it is present even for perfect spectra and dominates

at DESI densities. For DESI DR1, $\sigma_\delta \simeq 0.8$ per $0.8 \text{ \AA}$ pixel ($\Delta\chi_{\text{pix}} \simeq 0.55 h^{-1}\text{Mpc}$ at $z = 2.4$) gives $P_N \simeq 0.33 h^{-1}\text{Mpc}$, to be compared with $P_{1D}(k_\parallel = 0.1) \simeq 0.19 h^{-1}\text{Mpc}$; the $P_N \simeq 0.17$ case corresponds to the cleaned sample of Shaw et al. [5]. About 45 forests per deg^2 each covering $\sim 350 h^{-1}\text{Mpc}$ of a $712 h^{-1}\text{Mpc}$ deep slab give an unweighted available-sightline density $\bar{n}_{\text{los}} \simeq 22 \text{ deg}^{-2}$ at any one redshift (this replaces the 25 used in an earlier draft; the distinction from a noise-weighted effective density matters because the harmonic-mean weighting below already accounts for the noise); 50 and 100 deg^{-2} stand for a complete DESI or a denser survey. Two limitations of (26) should be kept in mind: replacing the full sampling covariance by independent Gaussian modes with this diagonal power ignores the mode couplings induced by sparse sampling, and Poisson sampling has no sharp Nyquist frequency, so the $l_{\text{max}} = \pi\sqrt{\bar{n}}$ cutoff is an additional restriction rather than part of the noise (relaxing it to $2\pi\sqrt{\bar{n}}$ lowers $N_\kappa(100)$ by 18 per cent). Both are to be validated against the discrete estimator on mocks.

Inverse-variance weighting of sightlines. Equation (26) with a single P_N describes a homogeneous population with equal weights. Real quasars span more than two magnitudes, so P_N varies by well over an order of magnitude from sightline to sightline. The discrete estimator (9) weights every pixel by $C^{-1} = (C^0 + N)^{-1}$ automatically; in the continuum formula the same C^{-1} weighting appears because independent sightlines contribute additively to the Fisher information per mode, so that

$$\frac{1}{\mathcal{N}(k_\parallel)} = \bar{n}_{\text{eff}} \left\langle \frac{1}{P_{N,a} + P_{1D}(k_\parallel)} \right\rangle_a, \quad (27)$$

a harmonic rather than arithmetic mean over the noise distribution. The aliasing term stays inside the average because the unresolved small-scale structure of each skewer is independent noise for that skewer. We model $P_{N,a}$ as log-normal with median $0.33 h^{-1}\text{Mpc}$ and ln-scatter σ_{ln} ; since $P_N \propto \sigma_\delta^2 \propto f^{-2}$ for background-limited spectra, a factor-of-three scatter in quasar flux gives $\sigma_{\text{ln}} \simeq 2$, and $\sigma_{\text{ln}} = 1$ is a conservative case. Here $\bar{n}_{\text{eff}}$ is the unweighted density $\bar{n}_{\text{los}}$. Relative to the equal-weight median-noise case, (27) lowers $\mathcal{N}$ by a factor 1.1 ($\sigma_{\text{ln}} = 1$) or 1.1 ($\sigma_{\text{ln}} = 2$) at $k_\parallel = 0.1 h \text{ Mpc}^{-1}$, and by 1.1 or 1.4 at $k_\parallel = 1 h \text{ Mpc}^{-1}$ where pixel noise dominates over aliasing. The actual DR1 distribution of per-pixel noise should replace this model; it is available from the public DR1 delta files. The quasar template has $\bar{n}_q = 25 \text{ deg}^{-2}$ in the slab (shot noise $1/\bar{n}_q = 1.2 \times 10^{-5} \text{ sr}$), the CMB noise is the white-noise equivalent of Section 3 recalibrated after extending the Limber integral for $C^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}}$ to recombination (the earlier $z = 6$ cap dropped 7–32 per cent of it), and the deprojected map has power $C^{\kappa_\perp\kappa_\perp} = C^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}} + N^{\text{CMB}} - 2\beta C^{q\kappa_{\text{CMB}}} + \beta^2(C^{qq} + 1/\bar{n}_q)$; for the matched template the same with $\beta = 1$, $q \rightarrow \hat{\kappa}_s$. Deprojection removes only $r_{q\kappa}^2 = 0.033$ of the CMB lensing power at $L = 100$, and the added shot-noise term is comparable to $C_{100}^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}} = 1.59 \times 10^{-7}$. The variance of the forest side includes the response sample variance, $C^{\hat{\kappa}\hat{\kappa}} = C^{\kappa_{\text{Ly}\alpha}\kappa_{\text{Ly}\alpha}} + 2aC^{\kappa_{\text{Ly}\alpha}\delta_s} + a^2C^{\delta_s\delta_s} + N_\kappa$ with $a = (R_\delta/2)R_{\kappa a}$; the factor 1/2 converts the power response $R_\delta = d \ln P_F / d\delta$ to the field modulation a to which the estimator responds.

4.4 Results

Table 2 and Figures 4–5 give the outcome. The S/N is quoted both over all L and over $40 \leq L \leq 300$, the range a cell- or pair-based measurement with a released CMB map will actually cover; the latter is the headline. Three points:

1. **The contamination is comparable to the signal.** With $R_\delta = 2$ the response term in $\langle \hat{\kappa} \kappa_{\text{CMB}} \rangle$ is 0.82 times the lensing signal at $L = 40$ and 1.63 times at $L = 100$ (an earlier draft quoted twice these values, having applied the power response R_δ where the field response $R_\delta/2$ belongs). The ratio grows with L because the slab density spectrum is shallower than the lensing spectrum, and the estimator transfers it almost fully, $R_{\kappa a} \simeq 0.8$ –1. The response term alone would be measured at 2.7σ over $40 \leq L \leq 300$ with DR1 noise: the deprojection is not optional, and the response measurement is a positive control.

Table 2: Reconstruction noise and cross-correlation signal-to-noise with realistic forest noise, for the slab $2.1 < z < 3.0$, $R_\delta = 2$, $b_q = 3.5$, $m = 0.5$, ACT-like CMB noise with $f_{\text{sky}} = 0.15$ and Planck-like with 0.22. “Naive” is $\langle \hat{\kappa} \kappa_{\text{CMB}} \rangle$, whose expectation is biased by the amount in the fifth column; “BH” is the per-screen hardened estimator of Section 2.7; “matched” is the deprojection with the kernel-matched quasar template (25); “response” is the significance of the response term itself. $\sigma_{\text{ln}} > 0$ rows use the inverse-variance weighting of Eq. (27).

$\bar{n}_{\text{los}}$	P_N	σ_{ln}	$N_\kappa(100)$	bias/signal at $L = 100$	cumulative S/N					
[deg $^{-2}$]	[$h^{-1}\text{Mpc}$]				all L , ACT			$40 \leq L \leq 300$		
					naive	BH	matched	matched ACT	matched Pl.	response
22	0.00	0	2.43×10^{-6}	1.58	4.2	2.6	3.8	3.1	3.2	5.8
22	0.17	0	6.73×10^{-6}	1.58	2.6	1.6	2.3	1.9	2.0	3.6
22	0.33	0	1.21×10^{-5}	1.59	1.9	1.2	1.7	1.4	1.5	2.7
50	0.00	0	5.94×10^{-7}	1.46	8.1	5.8	7.2	5.7	5.9	10.0
50	0.17	0	1.64×10^{-6}	1.44	5.2	3.8	4.6	3.7	3.9	6.4
50	0.33	0	2.89×10^{-6}	1.45	4.0	2.9	3.6	2.9	3.0	5.0
100	0.00	0	1.90×10^{-7}	1.37	12.8	10.8	11.2	8.4	8.7	14.1
100	0.17	0	5.32×10^{-7}	1.31	8.8	7.1	7.8	6.1	6.3	9.6
100	0.33	0	9.28×10^{-7}	1.32	6.9	5.5	6.1	4.8	5.0	7.6
22	0.33	1	1.09×10^{-5}	1.59	2.0	1.3	1.8	1.5	1.6	2.9
22	0.33	2	9.52×10^{-6}	1.60	2.2	1.4	1.9	1.6	1.7	3.1
50	0.33	1	2.59×10^{-6}	1.45	4.2	3.0	3.7	3.0	3.1	5.2
50	0.33	2	2.25×10^{-6}	1.46	4.5	3.2	4.0	3.2	3.4	5.6

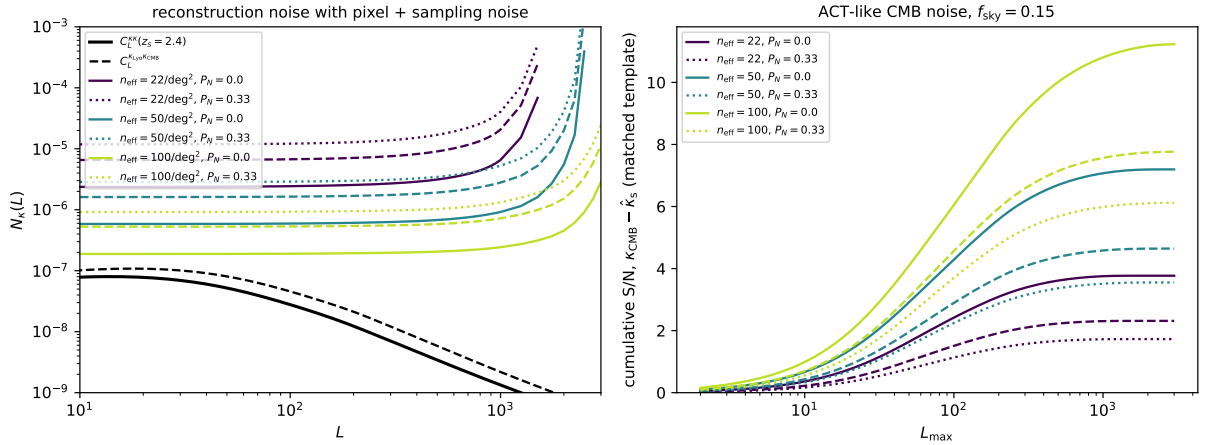


Figure 4: Left: reconstruction noise with the sampling and pixel noise of Eq. (26) (solid: perfect spectra, dotted: DESI DR1 pixel noise) for effective sightline densities of 25, 50 and 100 deg $^{-2}$. Right: cumulative signal-to-noise of the cross-correlation with the matched-template deprojected map, $\langle \hat{\kappa} (\kappa_{\text{CMB}} - \hat{\kappa}_s) \rangle$.

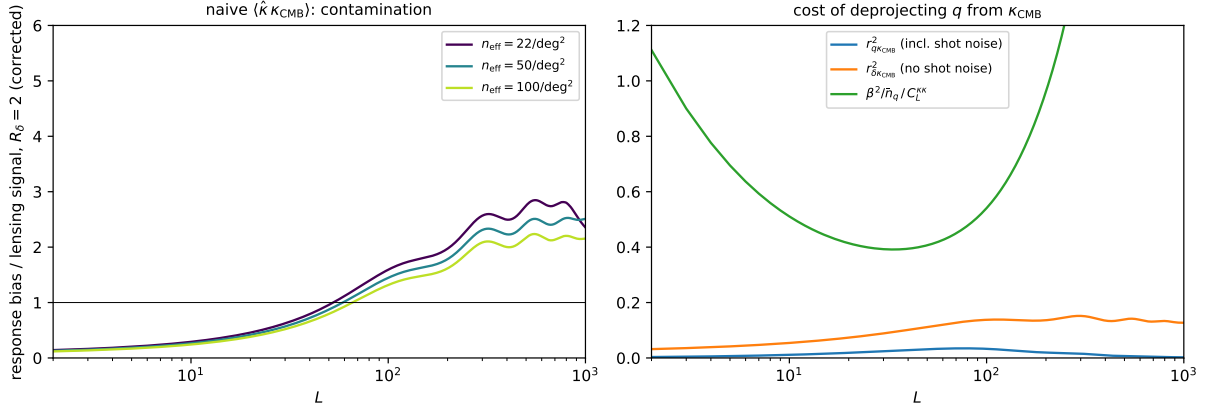


Figure 5: Left: ratio of the response contamination to the lensing signal in the naive $\langle \hat{\kappa} \kappa_{\text{CMB}} \rangle$ for a fiducial response $R_\delta = 2$, with DR1 noise. Right: the cost of deprojecting q from κ_{CMB} : the fraction of CMB lensing power removed ($r_{q\kappa}^2$) and the added quasar shot-noise term relative to $C_L^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}}$.

2. **Sampling noise, not pixel noise, sets the scale.** Aliasing alone raises $N_\kappa(100)$ from 2.5×10^{-8} (dense sampling, Section 3) to 2.43×10^{-6} at 22 deg^{-2} ; DR1 pixel noise raises it further to 1.21×10^{-5} . Only modes with $k \lesssim 0.15 h \text{ Mpc}^{-1}$ are signal dominated.
3. **What remains.** For DR1-like parameters the matched-template cross-correlation reaches $S/N = 1.4$ over $40 \leq L \leq 300$ (1.7 over all L) with ACT-like and 1.5 with Planck-like CMB noise, against 1.2 for the hardened estimator. Cleaner spectra ($P_N = 0.17$) give 1.9; inverse-variance weighting of a heterogeneous population with $\sigma_{\text{ln}} = 2$ gives 1.6; perfect spectra 3.1; a complete DESI with $\bar{n}_{\text{los}} = 50$ and clean spectra 3.7. Weighting recovers part of the gap to the perfect-spectra case but cannot cross it: the aliasing term $P_{1D}/\bar{n}$ is a floor, and it is the dominant noise at DESI densities. These numbers are consistent with the $S/N \simeq 3\text{--}4$ that Shaw et al. [5] find for the full DESI footprint against foreground galaxies, once scaled to their area.

The conclusion for DESI DR1 is a marginal measurement, below 2σ over the multipoles that will actually be used. The response term is measured at only 2.7σ by the same three-dimensional pair statistic, well below the 4.8σ of Karaçaylı et al. [14], because their per-skewer $P_{1D} \times \kappa_{\text{CMB}}$ statistic is immune to the sparse-sampling noise that dominates here; the two measurements are complementary, and the P_{1D} bispectrum can be used to fix the response amplitude externally. Reaching 5σ requires ~ 50 available sightlines per deg^2 at fixed redshift with cleaned spectra, i.e. the complete DESI survey or DESI-II. Low-redshift galaxies as the lensing tracer avoid the deprojection but not the sampling noise, so they offer no gain for the same forest data; their value is as an independent check. Not included, and all in the direction of lowering these numbers: forest non-Gaussianity (a factor 1.5–2.7 in Shaw et al. 4), continuum-fitting distortions, the noisier high-redshift end of the slab, the uncertainty in $b_q(z)$ and m , and the use of a white CMB noise level instead of the released N_L . These are Gaussian toy forecasts; the estimator of Section 5 and the injection tests described there are what will turn them into a measured significance.

5 The pair-template estimator

The cell-based statistic of the original plan (now superseded) had two avoidable losses: a top-hat cell window that suppresses the response near $L \sim 1/\theta_{\text{cell}}$, exactly where the signal is, and the squeezed-limit harmonic decomposition, whose error relative to the exact response is a factor $\text{sinc}(\mathbf{L} \cdot \boldsymbol{\theta}_{ab}/2)$ per pair, equal to 0.80 at $L = 300$ for a $30 h^{-1} \text{ Mpc}$ pair aligned with $\mathbf{L}$ (check 7 in

mathematica/pair_template_checks.wls). Both disappear if one uses the exact pair response of Eq. (5) with the CMB map itself as the template.

5.1 Estimator

For a deflection template $\alpha_T(\theta)$ and an amplitude A such that the true deflection is $A\alpha_T$, the response of the covariance of the pixel pair $i = (a, p)$, $j = (b, q)$ is, exactly to first order in α and for any L ,

$$R_{ij} \equiv \frac{\partial C_{ij}}{\partial A} = \nabla_{\theta} \xi(\theta_{ab}, \Delta\chi_{pq}) \cdot [\alpha_T(\theta_a) - \alpha_T(\theta_b)]. \quad (28)$$

With diagonal weights w_i (the picca inverse-variance weights, which already include the intrinsic forest variance) the quadratic estimator of (9) for this one parameter is

$$\hat{A} = \frac{1}{F} \left[\sum_{i < j} w_i w_j d_i d_j R_{ij} - b \right], \quad F = \sum_{i < j} w_i w_j R_{ij}^2, \quad b = \sum_{i < j} w_i w_j \xi_{ij} R_{ij}, \quad (29)$$

with $\langle \hat{A} \rangle = A$ by construction and $\text{Var}(\hat{A}) \simeq 1/F$ in the Gaussian, noise-dominated approximation; the actual variance comes from the null ensembles below. The mean field b is odd in θ_{ab} for isotropic ξ (check 4), so it nearly vanishes for an isotropic pair distribution, but it is computed exactly for the real geometry. Choosing $\alpha_T = \nabla \phi_T$ with $\phi_T(\mathbf{L}) = 2\kappa_T(\mathbf{L})/L^2$ and κ_T the deprojected, filtered CMB map, $\kappa_T(\mathbf{L}) = \kappa_{\perp}(\mathbf{L}) C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\perp}} / [(C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} + N_{\kappa}(L)) C_L^{\kappa_{\perp} \kappa_{\perp}}]$, makes $\hat{A}$ the Fisher-optimal amplitude of the cross-correlation within the diagonal-weight approximation, with the L -weighting done by the filter rather than by hand. Band powers follow from band-limiting κ_T ; each band has its own F_b . This is the cubic form $W_{ijk} \delta_F^i \delta_F^j \kappa_{\text{CMB}}^k$ of the task statement evaluated literally, with the exact response and no cells.

5.2 Compression

The deflection depends only on the sightline pair, not on the pixels, so all pixel sums can be done once:

$$\mathbf{V}_{ab} = \sum_{p \in a, q \in b} w_p w_q d_p d_q \nabla_{\theta} \xi_{pq}, \quad \mathbf{M}_{ab} = \sum_{p, q} w_p w_q \nabla_{\theta} \xi_{pq} \nabla_{\theta} \xi_{pq}^T, \quad \mathbf{B}_{ab} = \sum_{p, q} w_p w_q \xi_{pq} \nabla_{\theta} \xi_{pq}, \quad (30)$$

after which, for any template, with $\Delta\alpha_{ab} = \alpha_T(\theta_a) - \alpha_T(\theta_b)$,

$$\sum_{i < j} w_i w_j d_i d_j R_{ij} = \sum_{a < b} \mathbf{V}_{ab} \cdot \Delta\alpha_{ab}, \quad F = \sum_{a < b} \Delta\alpha_{ab}^T \mathbf{M}_{ab} \Delta\alpha_{ab}, \quad b = \sum_{a < b} \mathbf{B}_{ab} \cdot \Delta\alpha_{ab}. \quad (31)$$

For DESI DR1 there are $\sim 10^7$ sightline pairs within $30 h^{-1} \text{Mpc}$, so the pair-vector catalogue (seven numbers per pair) is $\sim 300 \text{ MB}$ and every template, band, jackknife region, simulation or injection costs 10^7 operations. The compression cannot be pushed to one vector per sightline: summing $\mathbf{V}_{ab}$ over an isotropic neighbourhood cancels at first order (check 3), because the information is the relative displacement along each pair direction, $\hat{\theta}_{ab} \cdot \Delta\alpha_{ab} \propto -\kappa - \gamma_+(\varphi_{ab})$ (checks 3b, 3c); the pair level is the right level.

5.3 Templates, nulls and calibration

- **Signal:** κ_T from $\kappa_{\perp} = \kappa_{\text{CMB}} - \hat{\kappa}_{\text{s}}$ with the kernel-matched quasar template (25), for ACT DR6 and Planck PR4 separately.
- **Response control:** κ_T from $\hat{\kappa}_{\text{s}}$ alone measures the response amplitude, the analogue of B_{FFq} , expected at $\sim 2.7\sigma$.

- **Curl null:** the template $\alpha_R = R_{90}\nabla\phi_T$ (the deflection rotated by 90°) has zero expected amplitude for any parity-invariant ensemble (check 6) and replaces the B -mode test of a cell-based shear field.
- **CMB null ensemble:** the 400 ACT (and Planck) simulated κ maps as templates give the distribution of $\hat{A}$ under no correlation, including the forest’s non-Gaussian variance, at 10^7 operations each. As the review notes, this is a conditional null: it does not vary the forest or the quasar template, so the covariance of the deprojected amplitude also needs jackknife resampling of the pair catalogue and joint mocks.
- **Injection calibration:** because lensing only remaps the transverse positions, shifting the observed sightline coordinates of the real data by $-A_{\text{inj}}\alpha_{\text{inj}}$ and recomputing (30) produces a data set with a known additional deflection. The recovered $\hat{A}_{\text{inj}}/A_{\text{inj}}$ calibrates F , including the effects of continuum-fitting distortion, weights, varying forest lengths and pair binning that the analytic $\nabla\xi$ cannot capture (review item 14): the response is purely geometric, so the calibration is exact for the geometric part and any approximation in the $\nabla\xi$ used in the weights affects only optimality. Each injection is one pair pass.
- **Continuum projection:** the projection P acts along each line of sight and ∇_θ transversally, so $PC_{,\kappa}P^T = \nabla_\theta(PCP^T)$ for a fixed projector (check 5); with varying forest lengths and weights this holds only approximately, which is what the injection calibration absorbs.

5.4 Relation to previous estimators and cost

Equation (29) is the estimator of Metcalf et al. [3] with their many Legendre coefficients of ϕ replaced by one or a few template amplitudes, which removes the ill-conditioned Fisher inversion that limited their maps; and it is the continuum estimator of Section 2.6 with the exact discrete geometry and C^{-1} approximated by inverse-variance pixel weights. The remaining suboptimality is the neglect of correlated forest signal between neighbouring sightlines in the weights; applying $(C^0 + N)^{-1}$ by conjugate gradient costs a pair pass per iteration and is expected to gain under 20 per cent in the noise-dominated regime. The one pair pass over $\sim 1.4 \times 10^{11}$ unique pixel pairs has to be benchmarked rather than estimated (review item 18): the arithmetic bound is minutes on 24 threads, realistic kernels with interpolation and scattered memory access are 10–50 times slower.

6 Caveats and next steps

- **Single source plane.** Pixels at different χ are lensed by slightly different convergences; the estimator as written recovers a $k_\parallel$ -weighted average. Modelling this is straightforward within (9) (each screen gets its own W_s).
- **Gaussianity.** The forest is strongly non-Gaussian on the scales that carry the information. Shaw et al. [4] find a factor 1.5–2.7 loss in signal-to-noise; the variance (12) will need the forest trispectrum or simulations.
- **Clustering bias.** The dominant systematic is the intrinsic $\langle\delta_F\delta_F\kappa_{\text{CMB}}\rangle$ bispectrum. The plan is to compute its projection onto $\hat{\kappa}$ using the measured responses of Chiang et al. [16] and the DESI Y1 measurement of Karaçaylı et al. [14], and to design a hardened estimator that is orthogonal to both the density and tidal responses; the price is the factor ~ 8 in noise found above for the density part alone.
- **Data-level effects.** Continuum fitting removes low- $k_\parallel$ modes and distorts the measured ξ ; DLAs, metals and quasar magnification bias [17] add non-lensing structure; the CMB maps carry tSZ/CIB residuals correlated with the forest’s foregrounds.

- **Numerics to firm up.** Replace the white CMB noise by the public ACT DR6 and Planck PR4 noise curves; refine the Arinyo parameters to the published $z = 2.4$ values; add pixel noise and a realistic $n(z)$ of forests.
- **Pending adversarial review.** The derivations in Sections 2.2–2.6 and the numbers in Section 3 are queued for review with codex/gpt-6-astra, which was unavailable when they were written.

A Symbolic checks

The scripts in `mathematica/` verify: (A) the second-order squeezed expansion (6), its angular average $2\mathcal{P} + l\mathcal{P}'$, its $\cos 2\varphi$ component $l\mathcal{P}'$, and the power-law result $(2 + n)\mathcal{P}$; (B) that the exact power spectrum of a field dilated and sheared by constant $\kappa, \gamma_1, \gamma_2$ agrees with (6) to first order, and that $\kappa = -\nabla^2\phi/2$ for the potential used; (C) that the real-space pair-displacement covariance (5), Fourier transformed in a periodic one-dimensional toy model, reproduces the mode coupling (4) for all tested (L, l) and vanishes for $l + l' \neq L$. `code/recon.noise.py` contains the white-spectrum mode-counting check of (15).

B Response to the first adversarial review

The codex review of 2026-09-10 (`report/reviews/codex_review.1.md`) confirmed the continuum normalisation and mode counting of Section 2.6 and raised 18 items. Changes made in response: the potential-to-convergence conversion and the mask mixing caveat in Section 2.5 (item 1); the removal of the incorrect $N^{(1)}$ -type argument (item 2); the factor 1/2 between the power response R_δ and the field modulation, which halves the contamination estimates (item 3); the kernel-matching condition (24) and the matched template (25), replacing the single slab-averaged coefficient (item 4); the consistent magnification treatment and the first-order nature of coefficient errors (items 5, 6); the withdrawal of the “quasars behind the slab” remedy in favour of tomographic sub-slabs (item 7); the corrected deprojected variance including the response sample variance (item 8); the cutoff-sensitivity and mode-coupling caveats of the noise model (item 9); the sightline density $\bar{n}_{\text{los}} = 22$ and the distinction from a weighted effective density (item 10); the Limber integral for $C^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}}$ extended to recombination with the CMB noise recalibrated (item 11); the $40 \leq L \leq 300$ headline range (item 12); the replacement of the cell-based estimator by the exact pair-template estimator of Section 5 (items 13, 15); injection calibration of the response through the continuum projection (item 14); and the revised validation plan in `PLAN.md` (items 16–18). Not yet done: the higher-order lensing corrections, a joint likelihood forecast with $b_q(z)$ and m free, validation of the sparse-sampling noise model against the discrete estimator, and the benchmark of the pair kernel.

References

- [1] Rupert A. C. Croft, Alessandro Romeo, and R. Benton Metcalf. Weak lensing of the Lyman α forest. *MNRAS*, 477(2):1814–1821, June 2018. doi: 10.1093/mnras/sty650.
- [2] R. Benton Metcalf, Rupert A. C. Croft, and Alessandro Romeo. Noise estimates for measurements of weak lensing from the Ly α forest. *MNRAS*, 477(2):2841–2847, June 2018. doi: 10.1093/mnras/sty806.
- [3] R. Benton Metcalf, Nicolas Tessore, and Rupert A. C. Croft. Reconstructing the gravitational lensing potential from the Lyman- α forest. *A&A*, 642:A122, October 2020. doi: 10.1051/0004-6361/202038056.

- [4] Patrick Shaw, Rupert A. C. Croft, and R. Benton Metcalf. Weak lensing the non-linear Ly α forest. *MNRAS*, 519(4):5236–5245, March 2023. doi: 10.1093/mnras/stac3786.
- [5] Patrick Shaw, Rupert A. C. Croft, and R. Benton Metcalf. Forecasting the Detection of Lyman-alpha Forest Weak Lensing from the Dark Energy Spectroscopic Instrument and Other Future Surveys. *The Open Journal of Astrophysics*, 8:10, January 2025. doi: 10.33232/001c.129066.
- [6] Frank J. Qu, Blake D. Sherwin, Mathew S. Madhavacheril, Dongwon Han, Kevin T. Crowley, Irene Abril-Cabezas, et al. The Atacama Cosmology Telescope: A Measurement of the DR6 CMB Lensing Power Spectrum and Its Implications for Structure Growth. *ApJ*, 962(2):112, February 2024. doi: 10.3847/1538-4357/acfe06.
- [7] Mathew S. Madhavacheril, Frank J. Qu, Blake D. Sherwin, Niall MacCrann, Yaqiong Li, Irene Abril-Cabezas, et al. The Atacama Cosmology Telescope: DR6 Gravitational Lensing Map and Cosmological Parameters. *ApJ*, 962(2):113, February 2024. doi: 10.3847/1538-4357/acff5f.
- [8] Julien Carron, Mark Mirmelstein, and Antony Lewis. CMB lensing from Planck PR4 maps. *JCAP*, 2022(9):039, September 2022. doi: 10.1088/1475-7516/2022/09/039.
- [9] A. G. Adame, J. Aguilar, S. Ahlen, S. Alam, D. M. Alexander, M. Alvarez, et al. DESI 2024 IV: Baryon Acoustic Oscillations from the Lyman alpha forest. *JCAP*, 2025(1):124, January 2025. doi: 10.1088/1475-7516/2025/01/124.
- [10] Alberto Vallinotto, Sudeep Das, David N. Spergel, and Matteo Viel. Lenses in the Forest: Cross Correlation of the Lyman- α Flux with Cosmic Microwave Background Lensing. *Phys. Rev. Lett.*, 103(9):091304, August 2009. doi: 10.1103/PhysRevLett.103.091304.
- [11] Alberto Vallinotto, Matteo Viel, Sudeep Das, and David N. Spergel. Cross-correlations of the Ly α Forest with Weak-lensing Convergence. Analytical Estimates of Signal-to-noise Ratio and Implications for Neutrino Mass and Dark Energy. *ApJ*, 735(1):38, July 2011. doi: 10.1088/0004-637X/735/1/38.
- [12] Cyrille Doux, Emmanuel Schaan, Eric Aubourg, Ken Ganga, Khee-Gan Lee, David N. Spergel, et al. First detection of cosmic microwave background lensing and Lyman- α forest bispectrum. *Phys. Rev. D*, 94(10):103506, November 2016. doi: 10.1103/PhysRevD.94.103506.
- [13] Chi-Ting Chiang and Anže Slosar. The Lyman- α power spectrum—CMB lensing convergence cross-correlation. *JCAP*, 2018(1):012, January 2018. doi: 10.1088/1475-7516/2018/01/012.
- [14] Naim Göksel Karaçaylı, Paul Martini, David H. Weinberg, Simone Ferraro, Roger de Bel-sunce, J. Aguilar, et al. CMB lensing and Ly α forest cross bispectrum from DESI’s first-year quasar sample. *Phys. Rev. D*, 110(6):063505, September 2024. doi: 10.1103/PhysRevD.110.063505.
- [15] Adrien La Posta and Emmanuel Schaan. Two shadows on two backlights: forecasting the Ly- α forest $\times$ CMB Lensing bispectrum from ACT/SO/S4 & DESI. *arXiv e-prints*, art. arXiv:2405.01628, May 2024. doi: 10.48550/arXiv.2405.01628.
- [16] Chi-Ting Chiang, Agnieszka M. Cieplak, Fabian Schmidt, and Anže Slosar. Response approach to the squeezed-limit bispectrum: application to the correlation of quasar and Lyman- α forest power spectrum. *JCAP*, 2017(6):022, June 2017. doi: 10.1088/1475-7516/2017/06/022.

- [17] Marilena Loverde, Stefanos Marnierides, Lam Hui, Brice Ménard, and Adam Lidz. Gravitational lensing as signal and noise in Lyman- α forest measurements. *Phys. Rev. D*, 82(10):103507, November 2010. doi: 10.1103/PhysRevD.82.103507.
- [18] Wayne Hu. Mapping the Dark Matter through the Cosmic Microwave Background Damping Tail. *ApJL*, 557(2):L79–L83, August 2001. doi: 10.1086/323253.
- [19] Wayne Hu and Takemi Okamoto. Mass Reconstruction with Cosmic Microwave Background Polarization. *ApJ*, 574(2):566–574, August 2002. doi: 10.1086/341110.
- [20] Oliver Zahn and Matias Zaldarriaga. Lensing Reconstruction Using Redshifted 21 Centimeter Fluctuations. *ApJ*, 653(2):922–935, December 2006. doi: 10.1086/508916.
- [21] Tingting Lu and Ue-Li Pen. Precision of diffuse 21-cm lensing. *MNRAS*, 388(4):1819–1830, August 2008. doi: 10.1111/j.1365-2966.2008.13524.x.
- [22] A. Pourtsidou and R. Benton Metcalf. Weak lensing with 21 cm intensity mapping at $z \sim 2-3$. *MNRAS*, 439(1):L36–L40, March 2014. doi: 10.1093/mnras/slt175.
- [23] A. Pourtsidou and R. Benton Metcalf. Gravitational lensing of cosmological 21 cm emission. *MNRAS*, 448(3):2368–2383, April 2015. doi: 10.1093/mnras/stv102.
- [24] Simon Foreman, P. Daniel Meerburg, Alexander van Engelen, and Joel Meyers. Lensing reconstruction from line intensity maps: the impact of gravitational nonlinearity. *JCAP*, 2018(7):046, July 2018. doi: 10.1088/1475-7516/2018/07/046.
- [25] Emmanuel Schaan, Simone Ferraro, and David N. Spergel. Weak lensing of intensity mapping: The cosmic infrared background. *Phys. Rev. D*, 97(12):123539, June 2018. doi: 10.1103/PhysRevD.97.123539.
- [26] Emmanuel Schaan and Simone Ferraro. Foreground-Immune Cosmic Microwave Background Lensing with Shear-Only Reconstruction. *Phys. Rev. Lett.*, 122(18):181301, May 2019. doi: 10.1103/PhysRevLett.122.181301.
- [27] Toshiya Namikawa, Duncan Hanson, and Ryuichi Takahashi. Bias-hardened CMB lensing. *MNRAS*, 431(1):609–620, May 2013. doi: 10.1093/mnras/stt195.
- [28] Noah Sailer, Emmanuel Schaan, and Simone Ferraro. Lower bias, lower noise CMB lensing with foreground-hardened estimators. *Phys. Rev. D*, 102(6):063517, September 2020. doi: 10.1103/PhysRevD.102.063517.
- [29] Abhishek S. Maniyar, Yacine Ali-Haïmoud, Julien Carron, Antony Lewis, and Mathew S. Madhavacheril. Quadratic estimators for CMB weak lensing. *Phys. Rev. D*, 103(8):083524, April 2021. doi: 10.1103/PhysRevD.103.083524.
- [30] Brandon Buncher, Gilbert Holder, and Selim Hotinli. Gravitational Lensing of Galaxy Clustering. *The Open Journal of Astrophysics*, 8:30, March 2025. doi: 10.33232/001c.132332.
- [31] Andreu Arinyo-i-Prats, Jordi Miralda-Escudé, Matteo Viel, and Renyue Cen. The non-linear power spectrum of the Lyman alpha forest. *JCAP*, 2015(12):017–017, December 2015. doi: 10.1088/1475-7516/2015/12/017.